

Estimating the size of informal economy in a post-transition country – the case of Poland

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ABSTRACT

The size of the informal economy in Poland is estimated by means of the Currency Demand Approach (CDA). Using quarterly data for the period 1999–2019, we adopt two separate econometric approaches. First, we specify a single equation model to estimate it with the Fully-Modified OLS method. Second, the CDA coefficients are treated as a cointegrating vector in a cointegrated VAR. The size of the informal economy in Poland is found to have diminished from about 32% of GDP in 2000 to about 12% of GDP in 2019. We provide confidence intervals for our estimates which, to our best knowledge, are rarely presented in the literature; their width ranges from 3 to 7% of GDP.

ARTICLE HISTORY

Received 3 November 2022
Accepted 19 June 2023

KEYWORDS

informal economy; currency demand approach; cointegration; vector error correction

JEL CLASSIFICATION

O17; H26; E26; C10

1. Introduction

The informal economy is a non-negligible phenomenon not only in developing, but also in transition or post-transition countries, including those in Central and Eastern Europe (CEE). It has many causes and consequences which are of great interest to policymakers and economists. The negative ones include reduced tax revenue, lower quantity and quality of public goods, deterioration of economic and social institutions and – as a result – lower economic growth. Consequently, policymakers are interested in reducing the size of informal economy. However, they have to be provided with the estimates of its size and the related estimates of various policy impacts. This is of particular importance in CEE countries, which are characterized by a larger size of informal economy in comparison to developed countries (Medina & Schneider, 2018).

Although the literature on measuring the informal economy is well developed, our knowledge on the size of this economy in CEE countries remains rather limited. First, most of the literature focuses on cross-country studies (Medina & Schneider, 2018) and thus does not scrutinize the specificity of individual country cases nor heterogeneity within panels. Second, the literature often uses the Multiple Indicators Multiple Causes (MIMIC) model as a macroeconomic estimation technique for the size of the informal economy, suffering from non-negligible algebraic identification deficiencies (Breusch,

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2016; Dybka et al., 2019; Feige, 2016a, 2016b). Third, alternative estimates of the size of the informal economy obtained from statistical offices in CEE countries are difficult to compare due to the limited availability of methodological descriptions (see Central Statistical Office, 2019, as an example). Moreover, these estimates are published with a substantial time lag. Furthermore, significant discrepancies exist between the estimates obtained from statistical offices and the literature. All these issues also affect Poland, the largest post-transition economy in the CEE region.

This paper contributes to the literature in two ways. First, we attempt to overcome the shortcomings listed above by using a theory-based method to estimate the size of informal economy in Poland. We use Currency Demand Approach (Cagan, 1958; Tanzi, 1983), with the currency equation estimated in two variants. In the first, we estimate a single equation model using the Fully-Modified OLS estimator. In the second, we modify the currency demand approach (CDA) treating the CDA coefficients as a cointegrating vector, estimated within a vector error correction model (VECM). In each case, we use quarterly data for the period 1999Q1 to 2019Q4 – almost a decade longer than in the last time series study on measuring the informal economy in Poland (Cichocki, 2009). Since CDA takes the demand for currency in circulation as the dependent variable, we decided to cut off the observations from 2020Q1 onwards as critically impacted by the COVID-19 pandemic (e.g. by changing consumption patterns or an abrupt switch towards safer, contactless payments).

We also contribute to the literature by providing confidence intervals for our estimates, something which is rarely done in the literature (Dybka et al., 2022; Mauleón & Sardà, 2017; Medina & Schneider, 2018).

Our results are of value not only for economists but also for policymakers, as they should allow them to evaluate if measures undertaken to reduce the size of informal economy are effective. The remaining part of this paper is structured as follows: in the next section, we discuss the relevant literature. We present our data and methods used in Section 3. In Section 4, we discuss our results. We conclude in the last section.

2. Literature review

2.1. *Methods of measuring the shadow economy*

The literature on the informal economy is very well developed, taking into account the fact how difficult this phenomenon is to capture. It focuses on such aspects as definitions of informal economy¹ (Koufopoulou et al., 2019; OECD, 2002) and the methods of its measurement (Adair, 2020; Cagan, 1958; Dybka et al., 2019; Frey & Weck-Hannemann, 1984; Gutmann, 1977; Tanzi, 1983; Kaufmann and Kaliberda, 1996; Medina & Schneider, 2018; Schneider & Enste, 2000). It also investigates the causes and consequences of the informal economy (Elgin & Erturk, 2019; Goel et al., 2017; Schneider & Enste, 2000; Tanzi, 2002). A separate strand focuses on the relation between the informal economy and the labour market (Bosch & Maloney, 2010; Schneider, 2014; Williams & Horodnic, 2015; Williams & Lansky, 2013).

As regards the measurement methods, early contributions can be dated back to Cagan (1958) and Macesich (1962). Both authors analysed changes in the ratio of cash to a broader money aggregate for the US and Canada respectively, noticing that such changes may reflect changes in the size of the informal economy. They discussed

potential determinants of this ratio and – using a very simple currency demand equation – pointed to tax rates as a key factor influencing the informal economy. Their approach was further developed by Gutmann (1977) who assumed that cash is the only means of payment used in the informal economy and that a base period existed in the past in which the size of the informal economy was equal to zero. He assumed further that the ratio of cash to deposits from this period would remain constant over time. As this ratio increased, Gutmann (1977) attributed the ‘excessive’ cash to the existence of the informal economy. Using this measure of excessive cash and assuming the same velocity of money in the formal and informal economy, he then calculated the size of the latter one.

About the same time, Feige (1979) proposed his transaction method taking as a starting point Fisher’s equation on the quantity of money. Assuming the existence of a base period without informal economy, he calculated the ratio of the supply of money to income for this period. Next, using this ratio and the supply of money for a given year, he calculated total income. Feige (1979) attributed the difference between this total income and the official income to the informal economy. He then assumed the same velocity of money in the formal and informal economy and calculated the size of the size of the informal economy.

This Currency Demand Approach was further developed by Tanzi (1983). He estimated a simple regression model with the ratio of cash to M2 as the dependent variable and several control variables, including the level of taxation. Based on the regression model, he calculated the total cash demand. Next, he estimated the same model but with the tax variable set to zero and calculated the cash demand. Tanzi (1983) attributed the difference between the total cash demand and the cash demand from the model without the tax variable to the informal economy. Assuming the same velocity of money in the formal and informal economy, he then obtained the size of the latter one.

Although fairly easy to apply, the CDA has drawbacks related to its assumptions, which can easily be contested (OECD, 2002; Thomas, 1999). First, the use of cash as the only means of payment in the informal economy, as well as the existence of a base period in which its size was equal to zero (Feige, 1979; Gutmann, 1977) or other anchoring techniques (Tanzi, 1983) are difficult to justify. Secondly, the assumption of equal velocity of money in the formal and informal economy is questionable. Thirdly, other factors than the informal economy can influence the cash to money aggregate ratio (Hanousek & Palda, 2006).

Additionally, a modern line of CDA critique (e.g. Marmora, 2021) suggests that the shadow economy can be highly exposed to the use of cryptocurrencies, i.e. a means of payment not included in the cash demand measures. This limitation applies predominantly to the most recent samples and will be of increasing importance in future research. One must bear in mind, however, that transactions performed with the blockchain technology are saved in the ledgers that pose risk of tracking (Adler, 2018), and high volatility of prices discourages from settling a substantial part of transactions using cryptocurrencies.

A second approach to measuring the size of the informal economy can be traced back to Frey and Weck (1983) and Frey and Weck-Hannemann (1984). They introduced the Multiple Indicators Multiple Causes (MIMIC) model, a macroeconomic estimation technique in which they assumed the size of the informal economy to be a latent variable. This variable can be estimated using various indicators of the informal economy (‘proxies’), such as the growth rate of the official GDP or the labour participation rate

for men (Frey & Weck-Hannemann, 1984). Furthermore, various causes of the informal economy, e.g. level of taxation, regulatory burden, and tax morality are also used in the estimation process.

The MIMIC model has subsequently been applied in a large body of papers (Bajada & Schneider, 2005; Chaudhuri et al., 2006; Dell'Anno & Schneider, 2003; Giles, 1999a, 1999b; Giles & Tedds, 2002; Hassan & Schneider, 2016; Medina & Schneider, 2018; Schneider, 2007; Schneider et al., 2010; Tafenau et al., 2010). However, this literature has faced a lot of criticism (Breusch, 2016; Feige, 2016a, 2016b; Helberger & Knepel, 1998; Hill, 2002; Kirchgässner, 2016; Smith, 2002). It concerns both the way the MIMIC framework has been presented, including the transparency of exposition and the replicability of results (Breusch, 2016; Feige, 2016a, 2016b) and the technicalities: specification, identification and estimation issues related with the MIMIC model (Breusch, 2016; Dybka et al., 2019).

Firstly, the transformation of variables to deviation from means and scaling of variables so that they have unit standard deviations which can lead to a biased predictor of the latent variable and which can make the whole procedure of estimating the informal economy sensitive to a change in units (Breusch, 2016). Secondly, mechanical differencing of variables to stationarity can at best lead to an efficiency loss (Breusch, 2016). The third issue is 'anchoring' of the index produced by the MIMIC model to external studies so as to obtain economically meaningful results. This in turn implies the variance of the informal economy, which, if too high, can lead to negative estimates of the informal economy (Dybka et al., 2019). In order to avoid such a problem, many authors add an arbitrary constant in the course of transformation. As a result, the MIMIC model has been shown to lead to questionable estimates (Breusch, 2016; Dybka et al., 2019; Feige, 2016a, 2016b). Furthermore, the literature using MIMIC models rarely provides confidence intervals for the estimates with Mauleón and Sardà (2017), Medina and Schneider (2018) and Dybka et al. (2022) being noteworthy exceptions.

Two other methods used for estimating the size of informal economy should be mentioned. The first one is the 'electricity method' used by Lacko (2000) who estimated a regression model with household electricity consumption as the dependent variable. However, this method is not free from problems either. The main concerns are related to the assumptions that electricity can serve as a good proxy of economic activity and to the value of the elasticity of electricity used in relation to GDP (Feige & Urban, 2003; Hanousek & Palda, 2006; OECD, 2002).

The second method is the discrepancy method which relies on National Accounts statistics. This method uses the GDP calculated according to the expenditure approach and the income approach. The difference between the results of these two approaches is then assumed to reflect the size of informal economy (Madzarevic-Sujster & Mikulic, 2002).

2.2. Size of the shadow economy in central and Eastern Europe

In the case of the size of the informal economy in CEE, our knowledge remains quite limited. First, most of the literature focuses on cross-country studies (Friedman et al., 2000; Hassan & Schneider, 2016; Medina & Schneider, 2018; Schneider, 2007; Schneider et al., 2010). These studies demonstrate a materially bigger size of the informal economy for CEE countries than for the established market economies. This holds also

for the enlarged EU. While the size of the underground economy for EU-15 is estimated to be on average 15–16% of GDP, it is about 23–24% of GDP for the CEE countries (Medina & Schneider, 2018). This literature obviously does not take into account individual country specificities when estimating the size of the informal economy and individual country studies are scarce. Second, alternative estimates of the size of informal economy obtained from statistical offices in CEE countries are difficult to comment on as the methods used are rather deficiently described (Statistics Poland, 2022) and a replication of the obtained results is not possible.

Both these problems also hold in the case of Poland, the largest post-transition economy of the region. The first studies focusing on the size of the informal economy were conducted before the transition to a market economy started (Bednarski et al., 1988) but obviously their comparability with the analyses for Poland after the transformation is limited. In the case of literature focusing on the period after 1989, Poland was typically included in cross-country studies, which did not allow for a thorough investigation of the size of the informal economy, especially over time. An exception is Cichocki (2009) who estimates the size of the informal economy only in the case of Poland and finds that the informal economy diminished from about 40% of GDP in 1995 to about 10% of GDP in 2007.

In the case of official estimates obtained from Statistics Poland² (Statistics Poland, 2022), two issues are noteworthy. First, these estimates are published with a delay of several years. Secondly, the methods used have not been described precisely enough to comment on the results. Furthermore, there is a significant difference between these estimates and the estimates obtained in the literature (Medina & Schneider, 2018) as presented in Figure 1.

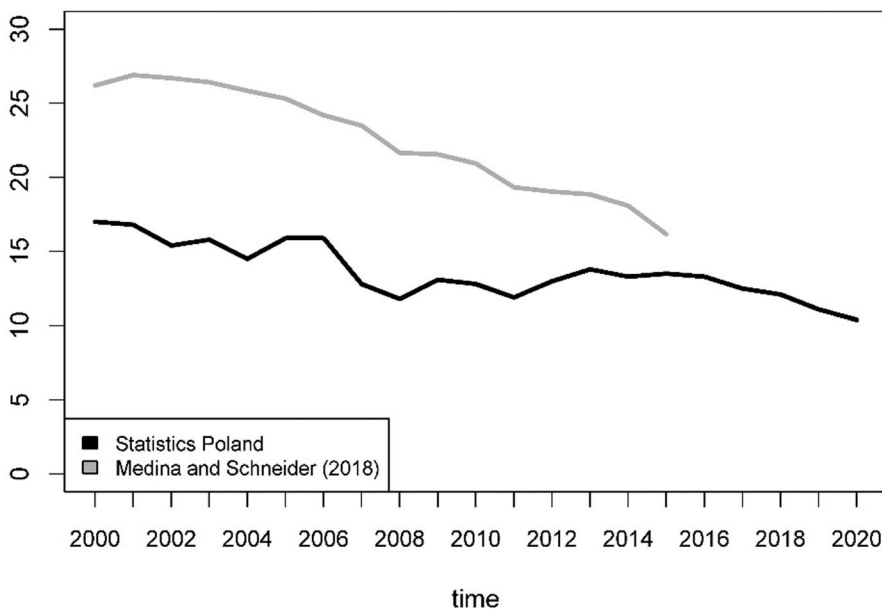


Figure 1. Estimates of the size of informal economy in Poland 2000–2020 (in % of GDP). Source: Medina and Schneider (2018), Central Statistical Office (2008, 2012, 2013, 2016), Statistics Poland (2019, 2020, 2022).

The estimates of the informal economy obtained from Statistics Poland are about 50% to 80% smaller than those of Medina and Schneider (2018) during most of the period from 2000 to 2015. Such discrepancies are difficult to explain as the methods used by Statistics Poland have been described in little detail.

All these shortcomings suggest that further studies on the size of informal economy in Poland are pertinent. Therefore we proceed with the estimation of this size using a version of the Currency Demand Approach.

3. Data and methods

In our empirical investigation, the dependent variable in the currency demand equation is the ratio of cash in circulation (outside monetary financial institutions) relative to the M1 monetary aggregate. In line with the CDA framework, we regress it on a number of variables representing the fundamental part of cash demand and the excess demand related to the existence of shadow economy that prefers cash as a means of transaction settlement, with Polish quarterly data from 1999Q4 to 2019Q4 (81 observations). We subsequently use the estimated models to decompose the total cash demand into respective contributions, and hence measure the shadow economy.

The measure of cash in circulation [in PLN] has been taken from the database of the National Bank of Poland. This is also the case for the denominator, the M1 monetary aggregate [in PLN]. The construction of the dependent variable follows Dybka et al. (2019) and generally puts into the foreground the problem of means of payment selection faced by the parties of a transaction, while removing the background information about the general value of transactions. Consequently, the variables reflecting a general value of transactions (like deflated measures of consumption or GDP) become redundant as explanatory variables. It should be noted that our data on cash does not include any measure of cryptocurrency stock. The holdings of cryptocurrencies cannot be credibly determined on a country level and based on available estimates, this market had negligible size in Poland until 2017, i.e. in the timespan covering 90% of our sample. At the endpoint of this sample, the ratio of global cryptocurrency market capitalization to global M1 (according to statista.com and Bank of International Settlement estimates) amounted to 1.3% (and near zero before 2014, e.g. throughout 75% of the sample). This appears as negligible as compared to the cash/M1 ratio values of 20.0–37.9% throughout the sample, and taking into account that most Poles who held cryptocurrencies around 2017 did that for speculative motives (Piech & Kacwin, 2017).

In line with the typical CDA setup, the set of explanatory variables in consideration includes two subsets.

Firstly, the variables assumed to reflect agents' motivation to use cash as a means of payment facilitating hiding transactions, thus rendering them informal transactions, include: (i) **direct taxes**: the ratio of PIT and CIT tax revenues (as reported in the *Statistical Bulletin* of Statistics Poland) to GDP (taken from the same source) [natural logarithm]³; (ii) **unemployment rate**: reported by Statistics Poland and Eurostat as the Labour Force Survey result (from 2000 to the end of the sample), and officially registered unemployment for 1999⁴; (iii) **government effectiveness**: the World Bank's institutional index, prepared as part of the Worldwide Governance Indicators project, related to the quality of public services.⁵

The second subset includes other structural determinants of the demand for transactional cash: (i) **electronic payments**: the ratio of payments settled with cards [PLN] to consumption of households and non-profit institutions serving households from the National Accounts data [PLN, current prices, seasonally adjusted, in natural logarithm]. This data is available at a quarterly frequency since 1999 from the National Bank of Poland and Eurostat. Unfortunately, this information is not publicly available in a potentially more insightful breakdown into individual sectors of retail trade; (ii) **nominal interest rate**: 3-month money market rate WIBOR [%], retrieved from the European Central Bank's database; (iii) **inflation**: consumer price index inflation [% y/y] from Statistics Poland.

It should be noted that some variables were excluded from the analysis, despite the fact that they had been originally considered as potential explanatory variables. The ratio of **indirect tax revenues** to GDP has been dropped from the model because its variance was heavily impacted by the institutional changes in the year 2016 (see Figure A1 in Appendix 3), related predominantly to combating tax frauds on registered transactions rather than to the informal economy, understood as unreported transactions, and hence not fitting into the CDA framework. The World Bank's index of **rule of law** has virtually turned out to be the mirror image of the government effectiveness variable, and hence we omitted it due to collinearity. The upward trends in **consumption**, or **GDP per capita in PPS**, did not contribute additional variation over the electronic payments variable.

We also considered to replace the unemployment rate with the rate of **unemployment and inactivity**, bearing in mind the relatively high share of inactive people in the population and the cyclical nature of unemployment itself; however, we eventually opted for the unemployment rate on the basis of the statistical performance in the modelling stage.

For all of the variables in question, data prior to 1999Q4 is also available. We decided, however, not to include this data in the sample due to the presence of the trends specific to the economic and political transition period: the reduction of unemployment rate, nominal interest rate and inflation rate, that would otherwise dominate the variance of the dataset.

For the sake of completeness in the exposition of the data sources, it should be noted that after the econometric modelling we use data on GDP [PLN, current prices, seasonally and working day adjusted] in subsequent calculations to obtain our estimates of the shadow economy as % of GDP.

We estimate the currency demand equation using two econometric approaches. The first one consists in applying the Fully-Modified OLS estimator (Phillips & Hansen, 1990) in the single equation framework:

$$y_t = \theta x_t + \epsilon_t, \quad (1)$$

where y – cash in circulation to M1, x – set of explanatory variables (direct taxes, unemployment rate, government effectiveness, electronic payments, nominal interest rate, inflation), including constant. As compared to OLS, this estimator takes account of the long-run correlation between the cointegrating equation (residuals) and the regressors that potentially face stochastic innovations. The second approach involves β as a cointegrating relationship in a Vector Error Correction Model (VECM; cf. Juselius, 2008) model,

using the Johansen approach:

$$\Delta \mathbf{z}_t = \alpha^T \beta \mathbf{z}_{t-1} + \sum_{i=1}^n \Gamma_i \Delta \mathbf{z}_{t-i} + \Gamma_0 \mathbf{d}_t + \varepsilon_t, \quad (2)$$

where $\mathbf{z}_t = [y_t \quad \mathbf{x}_t]$ and the first row of β conceptually corresponds to $[1 \quad -\theta]$ as denoted in (1).

The use of a VECM can be formally motivated with the non-stationarity of most of the variables in question. The key dependent variable, the log of cash in circulation outside MFI to M1 ratio, is found to be I(1) according to both ADF and KPSS tests. The same is the case for most of the explanatory variables. This group includes three clear I(1) cases (government effectiveness, inflation rate, nominal interest rate) and three borderline cases (with mixed test results, or results depending on the significance level; see [Table A7](#) in Appendix 3 for details). Within the latter group, the direct tax revenues deem as close to stationarity, while the electronic payments are I(1) in logged form and ADF test. A possible I(2)-ness of the unemployment rate is difficult to defend on the level of the economic reading and might be due to a relatively short sample. All in all, we decided to include all these variables into the cointegrating space in levels as one would handle I(1) variables.

Part of the data is characterized by seasonality. Since this is not the case for the dependent variable, we do not assume that this translates into the seasonality of cash intensity of M1, and – hence – the underlying notion of CDA approach excludes the impact of these seasonality patterns on the informal economy size. Consequently, we prefer to remove the seasonality to avoid introducing collinearity within the set of regressors, related to their seasonal patterns. The TRAMO-SEATS procedure was used to seasonally adjust three explanatory variables: direct tax revenues, unemployment rate and electronic payments (see [Figure 2](#)).

It should perhaps be mentioned that direct tax revenues represent the only case in which seasonal frequencies contribute to the series to a material extent. There are good reasons for this, going back to the tax system design and labour market institutions, but far from the shadow economy determination, i.e. a process that this variable is intended to represent in the model. The seasonal fluctuations stem from i.a. annual cycle of clearing the taxpayers' accounts in Q1 and Q2, increase of tax base due to cumulated payoff of variable wage components in Q4, taxpayers' continuous move towards higher marginal rates on a progressive scale towards the end of the year, and alike. By using the seasonally adjusted version of this variable, we assume that taxpayers do not flow in and out of the shadow sector in response to these seasonal, fully anticipated phenomena. As regards the other two variables, the results remain virtually unchanged whether or not one uses a seasonally adjusted version of the series.

Additionally, for the sake of residual normality, four dummy variables related to periods of financial market hypes or tensions have been added (2002q1, 2007q2, 2008q4, 2009q1).

4. Results and discussion

4.1. Statistical and economic model properties

[Table 1](#) contains the first set of estimates of the currency demand equation, obtained as Fully-Modified OLS regression estimates. In the light of the results reported in [Table A7](#),

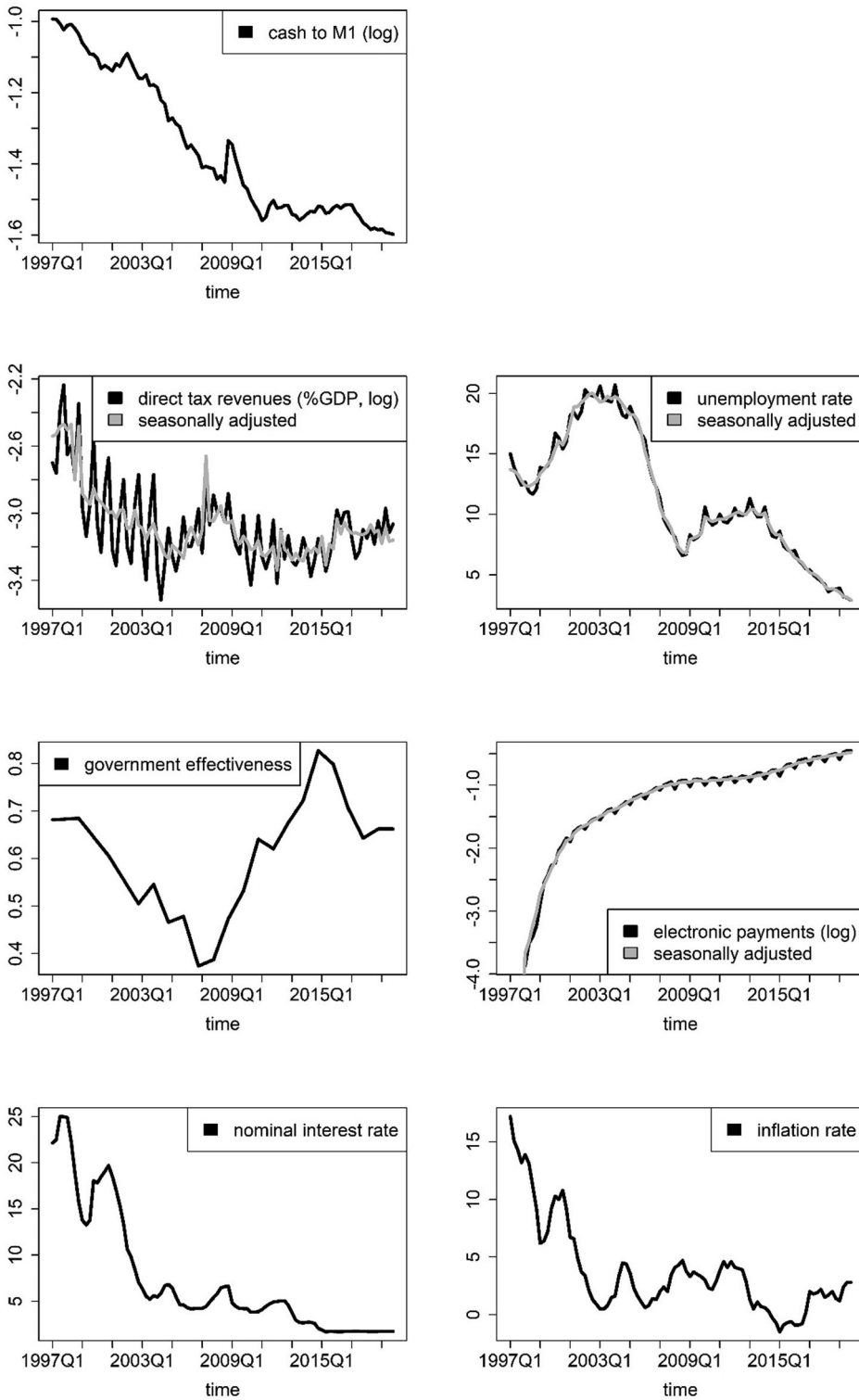


Figure 2. Main variables used in the CDA equation. Source: authors' elaboration.

Table 1. Currency demand model – FM-OLS, OLS, VECM-1 model ML and VECM-2 model ML estimates (1999q4-2019q4).

Variable	FM-OLS estimates				OLS estimates				VECM-1 model estimates			VECM-2 model estimates		
	Coefficient	Std. Error	t-Statistic	Prob.	Coefficient	Std. Error	t-Statistic	Prob.	Estimate	Std. Error	t-Statistic	Estimate	Std. Error	t-Statistic
Direct tax revenues	0.210	0.109	1.928	0.058	0.215	0.069	3.119	0.003	0.504	0.078	6.438	0.470	0.103	4.583
Unemployment rate	0.008	0.005	1.640	0.105	0.011	0.003	3.652	0.001	0.009	0.003	3.261	0.015	0.004	4.104
Gov. effectiveness	−0.224	0.091	−2.457	0.016	−0.156	0.058	−2.695	0.009	−0.398	0.072	−5.537	−0.181	0.101	−1.803
Electronic payments	−0.180	0.092	−1.952	0.055	−0.171	0.058	−2.928	0.005	−0.341	0.075	−4.566	−0.210	0.010	−2.098
Inflation	−0.018	0.008	−2.395	0.019	−0.012	0.005	−2.439	0.017	−0.007	0.006	−1.013	0.004	0.009	0.431
Nominal interest rate	0.009	0.007	1.294	0.200	0.008	0.004	1.772	0.080	−0.038	0.007	−5.517	−0.029	0.010	−2.995
Intercept	−0.893	0.322	−2.771	0.007	−0.953	0.204	−4.675	0.000	0.171			0.066		
Linear trend									0.000					

Note 1: For FM-OLS estimates – R -squared = 0.929, Adjusted R -squared = 0.924, S.E. of regression = 0.046. Long-run covariance estimate: prewhitening with 2 lags (selected with AIC under max. 4 lags), Bartlett kernel and Newey-West fixed bandwidth = 4. Note 2: For OLS estimates – R -squared = 0.939, Adjusted R -squared = 0.934, F -statistic = 190.462, Prob. (F -statistic) = 0.000, White test p -value (F) = 0.239, White test p -value (chi-sq) = 0.232, Breusch-Pagan p -value (F) = 0.180, Breusch-Pagan p -value (chi-sq) = 0.177. Source: authors' elaboration.

which can be found in Appendix 3, these estimates can be regarded as non-spurious provided that they constitute a cointegrating vector, i.e. that the implied linear combination of variables – and hence the residual vector – is stationary.⁶

In case of the FM-OLS and OLS estimates the regression problem exhibits a non-negligible degree of multicollinearity. The Variance Inflation Factors associated with 3 variables violate the rule-of-thumb limit of 10 (electronic payments – 27.1, nominal interest rate – 19.25, unemployment rate – 11.8). The economic reading of this result is neutral from the perspective of the CDA purpose, since all collinear variables characterize the structural determinants of cash demand, rather than the shadow economy. Hence the impact on the precision of shadow-related parameters, and the subsequent informal economy size calculation, can be regarded as limited.

Most of the estimated coefficients both in case of the FM-OLS results and the OLS results exhibit the expected sign. The ratio of direct tax revenues to GDP and the unemployment rate are positively related to our dependent variable, whereas the government effectiveness, electronic payments and inflation are negatively related. Rising unemployment and declining government effectiveness all boost the shadow activity, in line with the previous literature, such as the most recent international panel CDA estimates of Dybka et al. (2019, 2022). It is noteworthy that, unlike Dybka et al. (2022) and in line with earlier literature, we obtain a meaningful and intuitive estimate for direct tax revenues, which can be attributed to country-specific factors diluted when moving to a panel dataset.

There are, however, two striking results. First, the sign of the OLS estimate for the nominal interest rate coefficient is positive, although its size is quite small. The second is the insignificance of this variable in case of the FM-OLS estimates. The recent literature background in CDA models is mixed: the previously referred panels by Dybka et al. (2019, 2022) also contain a positive estimate, with a high inclusion probability of the nominal interest rate or a small inclusion probability and marginally positive estimate for the real interest rate. A possible explanation for such results is the fact that a rise in nominal interest rate may be related to a shift from current deposits into term deposits (decline in the denominator of the dependent variable – M1), with a weaker or similar shift from cash into term deposits. In the former case, the numerator of the dependent variable declines less than the denominator leading to a rise of the dependent variable whereas in the latter case it declines to such an extent that the dependent variable remains unchanged.

The second set of estimates has been obtained with VECM. CDA coefficients are retrieved as estimates of the cointegrating relation normalized with respect to the cash-to-M1 ratio. The total number of cointegrating relations, a central parameter for this model type, has been determined by the standard testing procedures (maximum eigenvalue and trace) of the size of the cointegration space (Juselius, 2008, chapter 8). According to the results, this number should equal 2 or 3, depending on the test and the significance level (see Table A4 in Appendix 1). Our decisive criterion is the sample size. For finite samples, a Bartlett correction can be applied (Johansen, 2002) to enhance the test size, and our effective sample size ($N=81$) largely corresponds to a range of 50–70 indicated by Juselius (2008, p. 141) as a typical use case where the Bartlett correction is substantial and decisive. The correction is conservative in terms of the suggested number of cointegrating relations, and clearly favours 2.⁷

Like with single equation methods, the VECM-1 model also exhibits the expected coefficient signs. The ratio of direct tax revenues to GDP and the unemployment rate are positively related to the ratio of currency in circulation to M1 whereas government effectiveness and electronic payments are negatively related. A noteworthy discrepancy is the fact that the sign for the coefficient related to the nominal interest rate is, contrary to the FM-OLS and the OLS results, negative. This would imply that a rise in the nominal interest rate lowers the ratio of cash in circulation to the M1 money aggregate.

The preliminary investigation of the isomorphic VAR models suggests the lag length of 3 (Schwarz and Hannan-Quinn information criteria), 6 (likelihood ratio test) or even 8 (Akaike IC, final prediction error and likelihood ratio test). Given this mixed evidence, and the similar inconclusiveness of the autocorrelation tests (LM and portmanteau), a lag length of 4, consistent with the quarterly seasonality cycle, has been selected as a safe-side solution.

In line with the Likelihood Ratio test indications (p -value = 0.65), we impose additional over-identifying restrictions: (i) exclusion of the deterministic trend from vector 1 (CDA), $\beta_{1,8} = 0$ (but a deterministic trend is apparently part of the second relationship); (ii) exogeneity assumptions with zero correction coefficients on vector 1 (for direct tax revenues, unemployment rate and government effectiveness) and on vector 2 (for unemployment rate, government effectiveness, inflation and nominal interest rate). The economic reading of these results is that tax revenues, unemployment and government effectiveness do not correct in response to a long-term misalignment from the CDA equation. In other words, there is no long-run feedback effect from cash demand to three shadow economy determinants.

The correction coefficient in the first equation (with the increment of log cash to M1 as dependent variable) for the first cointegrating relationship is -0.304537 with a t -statistic of -2.66 . This implies an almost identical half-life of disequilibrium as in the first approach under FM-OLS, and suggests that this relationship acts as a long-term attractor of the dependent variable interpretable as a cash demand equation for the CDA purposes. Table A1 in Appendix 1 presents complete estimates $\hat{\alpha}$.

One serious caveat should be emphasized as regards the economic interpretation of the VECM-1 model: two correction coefficients α are outside the range of non-explosive behaviour ($\hat{\alpha}_{1,6}^{VECM1} < -1$ and $\hat{\alpha}_{1,7}^{VECM2} < -1$). According to these results, inflation and nominal interest rate should be endogenous with respect to cash demand. It could be debated whether the choice of a specific money form (means of payment) impacts on the macroeconomic variables, but even if one permits this type of endogeneity to emerge (see Thoma, 1994, for an extensive discussion in terms of general money definition rather than cash share), an explosive model representation speaks against its credibility.

Hence, to avoid this and for the sake of robustness check, we propose a second VECM (see Table 1) that additionally imposes the exogeneity of both variables ($\alpha_{1,6}^{VECM2} = 0$ and $\alpha_{1,7}^{VECM2} = 0$). This leads to the rejection of the null in the LR test of restrictions (p -value = 0.000), but implies a stable behaviour and leaves the estimates largely unaffected on the qualitative level (see Table 1). All in all, being unable to demonstrate the superiority of VECM-1 over VECM-2, nor *vice versa*, we decided to use VECM-2-based CDA parameters to construct a separate version of the shadow economy estimate as a robustness check, i.e. to verify whether the seemingly minor

quantitative changes in cointegration coefficients β could have a material quantitative impact on the measurement. It should be noted that the error correction coefficient for cash to M1 on vector 1, $\hat{\alpha}_{1,1}^{VECM2} = -0.678$, i.e. the correction towards CDA equilibrium is stronger than in VECM-1. The coefficients in case of the VECM-2 model exhibit the expected signs and are in line with the signs of the coefficients obtained for the VECM-1 model.

4.2. Shadow economy measurement

Figure 3 presents the estimates of the informal economy size, computed for all estimated CDA coefficient vectors in a fairly standard procedure in the CDA literature, and consists in applying the estimated model to determine the shadow economy's contribution to the increase in cash/M1 ratio, and then translate this contribution into a fraction of GDP using some assumptions regarding the velocity of money. These include equality between the velocity in the official and in the shadow sector (see Dybka et al., 2022,

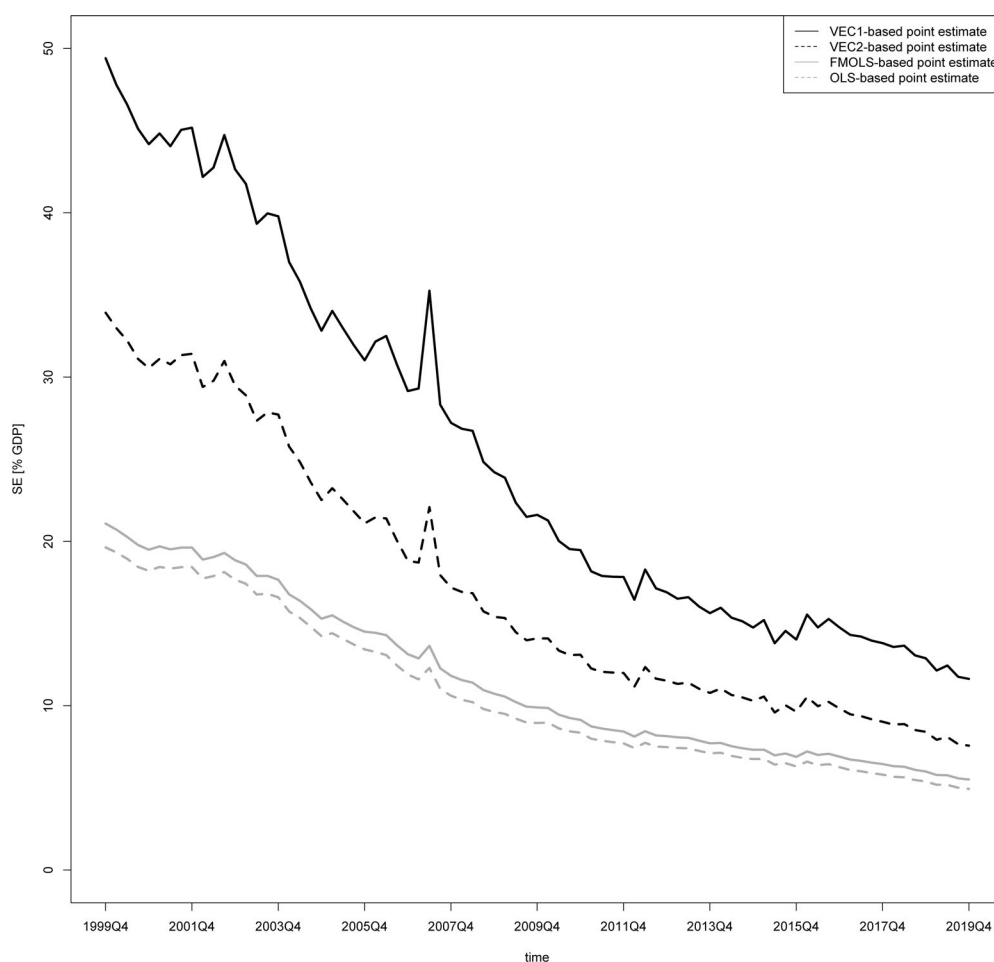


Figure 3. Informal economy as % of GDP: point estimates for Poland. Source: authors' elaboration.

for extended discussion) and using the time-averaged form of the computed velocity. The first assumption is just-identifying in the model, and difficult to verify with practically available data. The second is intended to smooth out short-term fluctuations, by which one implicitly regards both the money velocity and the shadow economy as persistent, long-term phenomena.

Technically, we first compute the theoretical value of log cash-to-M1 ratio, based on the respective equation ((i) FM-OLS estimated equation and (ii) cointegrating [equation \(1\)](#)). Second, we compute a counterfactual theoretical value, conditional on alternative values of the three variables related to the informal economy minimizing its prevalence. Then we compute the difference between the theoretical and the counterfactual theoretical value and the exponential function thereof (to revert the log transformation of the dependent variable). Next, we multiply the obtained value by the sample mean M1 to obtain the difference in the cash volume, attributable to the informal economy. Finally, we divide this cash volume by the sample mean velocity of money (cash to GDP) to generate the value of the informal transactions. This can be further expressed in % of GDP.

The counterfactual values from the second step shall be regarded as those at which the shadow economy would not exist. Despite a number of authors set these numbers, especially tax rates, at zero, we follow Dybka et al. (2019) who argue that this approach would be anti-conservative. At some positive, low tax rates risk-averse agents could enter the official market and gain the certainty of law compliance if inspected. We set this value at 1%, between the minimum value in the panel of World Bank's data on direct taxes to GDP ratio (0.26%) and the lowest value in the sample for Poland (3.5%). The unemployment rate is set equal to 0%, since 10 countries out of the Dybka et al. (2022) database exhibit values below 1%. The government effectiveness indicator value of 2.5 has been considered, the highest possible one by construction. In Subsection 4.3 we check for the sensitivity of these results to these assumptions.

Under both approaches, the informal economy exhibits a downwards trend over the period 2000-2019. Under the single-equation FM-OLS approach, its size declines from 20.1% in 2000 to 5.7% in 2019 (or, similarly, from 18.7% to 5.1% on the OLS basis). These estimates are substantially lower than both VECM-based estimates: VECM-2 coefficients imply a decline from 31.7% to 7.8% over the same period, while VECM-1 parameters go as far as to suggest a decline from 45.9% to 12.0%.

Notably, the VECM-based results are approximately 1.4–2.1 times higher than the single equation results towards the end of the sample period. This is attributable to the higher estimate of two semi-elasticities of the cash-to-M1: most of all, with respect to direct tax revenues, but also – to some extent – to the government effectiveness index. The estimates obtained using the VECM approach appear to be more in line with the bottom-up calculations by Statistics Poland: 12.0% of GDP in 2018 than the FM-OLS and OLS estimates which, as already stated, are substantially lower. Additionally, the findings obtained using the VECM approach appear to be better aligned with the estimates obtained by Cichocki (2009).

While the coefficient on direct tax revenues determines the bulk of the difference between the single-equation and multiple-equation approach, note that it is estimated with the highest standard error of all CDA parameters in all 4 underlying models. To

demonstrate the impact of this disturbing property, we construct confidence intervals around the point estimate of the shadow economy reflecting the statistical uncertainty around the CDA parameters. For the single-equation approach, we draw $S = 10\,000$ vectors from the multivariate normal distribution with the mean equal to the point estimates of the coefficients and variance-covariance matrix equal to the evaluation of the estimator variance-covariance matrix. For every $s = 1, \dots, S$ vector of parameters, we compute the informal economy size and then the respective quantiles over the resulting S -element vector: (i) quantiles of order 0.05 and 0.95 as the upper and lower limit of the 90% confidence interval, as well as (ii) quantiles of order 0.005 and 0.995 as the respective limits of the 99% CI. The result is illustrated in Figure 4.

The confidence interval for VECM approach reflects the uncertainty around the relevant estimates of the cointegrating relationship. The 3 relevant parameters ($\beta_{1,2}$, $\beta_{1,3}$, $\beta_{1,4}$) were drawn from normal distributions with means at point estimates and standard deviations equal to the standard errors in the respective VECM specification.

The confidence intervals shown in Figures 4 and 5 should be treated as rough approximations, since the true distributions for the estimators of the cointegrating vectors are not straightforward to obtain. The use of bootstrapping methods in such cases is not straightforward, either, especially for VECM-1, i.e. in the absence of stable correction mechanism. Note that, due to the functional form adopted (the dependent variable in natural logarithm), the confidence interval for the shadow economy in level (and ratio to GDP) is asymmetric, with a higher upside risk.

As a result, for the year 2019 (in terms of the average over quarters), the confidence intervals of 3.8–8.6% of GDP (99% confidence level) and 4.3–7.3% of GDP (90% confidence level) have been obtained with the FM-OLS-based measurement. The related uncertainty is much wider under VECM-based measurements: for the VECM-1 option, from 7.6% to 18.8% (99% confidence level) and from 9.0% to 16.0% (90% confidence level), and slightly more narrow for VECM-2 (99%: from 4.2% to 14.8%; 90%: from 5.3% to 11.6%). While the central path appears to be more reliable for the second approach (VECM) – as judging against the previous evidence for Poland – it must be kept in mind that it also yields higher uncertainty.

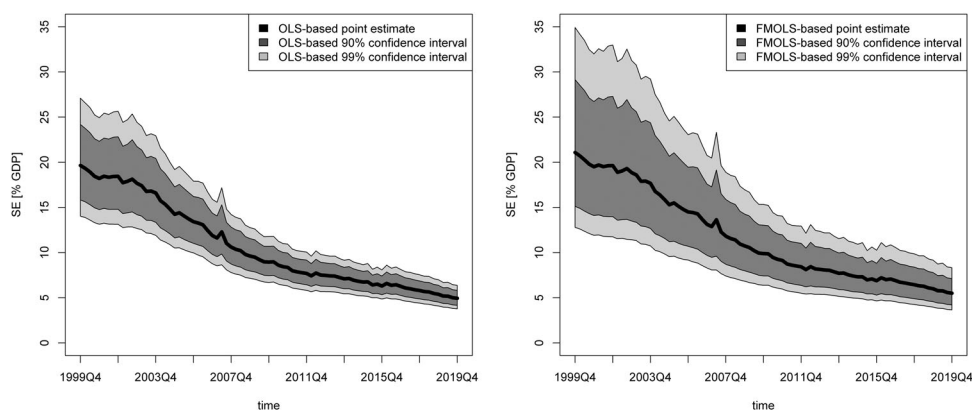


Figure 4. Confidence interval around the informal economy size: single-equation approaches (OLS, FM-OLS). Source: authors' elaboration.

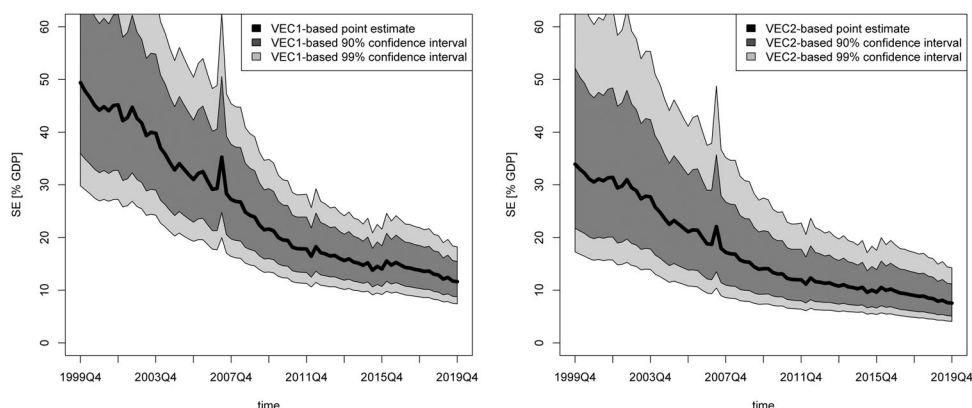


Figure 5. Confidence interval around the informal economy size: VECM approaches. Source: authors' elaboration.

As a robustness check, we look at the alternative choice of independent variable as cash to M2 ratio (see Appendix 1), concluding that the signs, significance and magnitudes of the coefficients for the informal economy determinants in all model versions remain largely unchanged.

4.3. Robustness check for no-shadow economy scenario

An inherent property of the CDA approach is the reliance on the assumption that the zero-shadow economy scenario corresponds to certain values of the shadow economy determinants. Recall that our assumptions involved 1% direct tax to GDP ratio, 0% unemployment rate and government effectiveness index equal to the maximum possible value of 2.5. To verify the sensitivity to these assumptions, we build 2 alternative scenarios. In the extreme scenario (i.e. yielding the highest shadow economy estimates), we set the direct tax to GDP to 0.26%, which is the minimum value in the World Bank's panel and the unemployment rate to 5% (roughly corresponding to the first quartile of panel unemployment rate in Dybka et al. (2022) dataset⁸). In the conservative scenario, we set the direct tax to GDP ratio to 3.5% (minimum for Poland) and the government effectiveness index to 2.43 (maximum over the reported panel).

For the sake of brevity, we only report the sensitivity check for FM-OLS and VECM-1 estimates. The result is mixed: much as the FM-OLS results remain relatively robust with respect to the adopted assumption, the VECM-1 results are not (see Figure 6). The deviations between scenarios decline along with the estimated shadow economy level (i.e. towards the end of the sample), amounting to -1.1 p.p. (conservative scenario) and $+2.3$ p.p. (extreme scenario) – a span covered by the previously reported confidence intervals. On the other hand, the VECM-1 estimates appear to be highly sensitive to scenario definitions, from -5.2 p.p. (conservative) to $+12.5$ p.p. (extreme).

The major contributor to these differences is the counterfactual direct tax rate and its accompanying coefficient, differing from 0.210 under FM-OLS to 0.504 under VEC-1. This reveals a potential shortcoming of the CDA approach, consisting in an arbitrary parametrization of the zero-shadow economy situation. A potential nonlinearity in the vicinity of

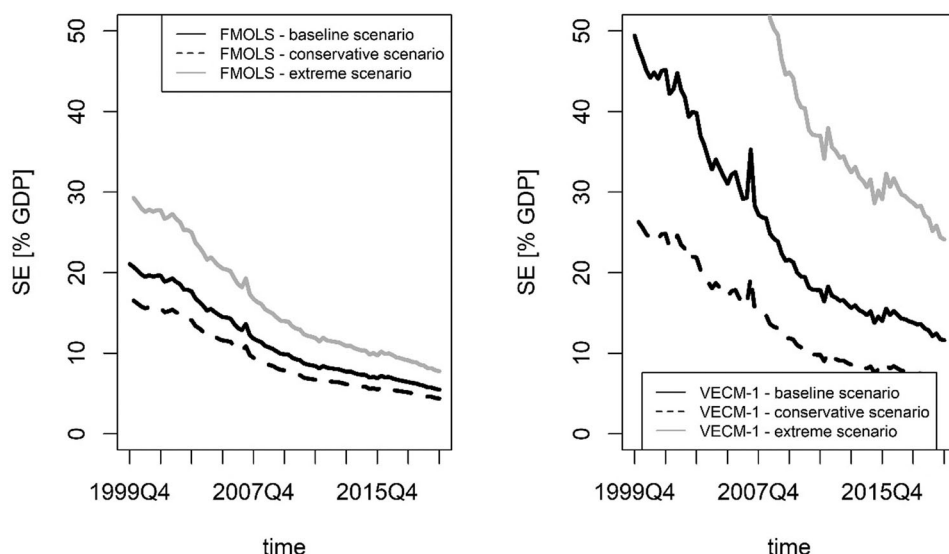


Figure 6. Sensitivity of the shadow economy estimates to the CDA assumptions of zero-shadow economy scenario. Baseline scenario: tax rate 1%, unemployment rate 0%, government effectiveness 2.5. Conservative scenario: tax rate 3.5%, unemployment rate 5%, government effectiveness 2.43. Extreme scenario: tax rate 0.26%, unemployment rate 0%, government effectiveness 2.5. Source: authors' elaboration.

zero, resulting in agents' choice to operate in the official market even in the presence of low positive tax rates, requires further micro-level econometric or behavioural studies. This nonlinearity cannot be determined a conventional macro-level econometric study like this, in which the well-explored part of the data domain is used to estimate parameters, but these parameters are then taken to emulate a borderline scenario of 'no-shadow economy'. This is especially true under the log-log specification, well-behaved with the real-world data but no more so when one of the variables is considered to fall near zero.

5. Conclusions

Our knowledge on the size of informal economy in Central and Eastern Europe countries remains limited because most of the literature focuses on cross-country studies (Medina & Schneider, 2018). Alternative estimates of the size of informal economy obtained from statistical offices in CEE countries are difficult to comment on as the methods used are rather poorly described and these estimates are published with a significant time lag (Statistics Poland, 2020). However, implementation and evaluation of policies aimed at reducing the informal economy demands that policymakers be provided with credible estimates of the informal economy.

In this paper, we attempt to overcome these shortcomings for Poland – the largest post-transition economy in the CEE region. We use quarterly data for the period 1999Q4 to 2019Q4 and apply the Currency Demand Approach. First, we estimate a single equation model with a Fully-Modified OLS estimator. Then we modify the CDA,

treating the CDA coefficients as a cointegrating vector, estimated within a Vector Error Correction Model (VECM). Our results show a decreasing size of the informal economy for Poland in the period 2000–2019. It exceeded 30% of GDP in 2000 when using the VECM and, at the same time, accounted for about 18.7–20.1% of GDP when applying the single equation model. This size diminished in 2019 to about 7.8–12.0% of GDP according to VECM and about 5.1–5.7% of GDP in the single equation models. The latter ones produce significantly lower estimates of the shadow economy than VECM which yields results in line with the estimates of Statistics Poland (2020) and earlier estimates focusing only on Poland (Cichocki, 2009).

We also contribute to the literature by providing confidence intervals for our estimates, something which is rarely done (Mauleón & Sardà, 2017; Medina & Schneider, 2018). In the case of the single equation model, the 90% confidence interval ranges from about 15% to 29% of GDP in 2000 and from about 4.3% to 7.3% of GDP in 2019. In case of the VECM, we obtain a much higher uncertainty as the 90% confidence intervals range in 2019 from 9.0 to 16.0% of GDP or 5.3 to 11.6% of GDP (depending on the model version).

Our results are of value not only for economists, but also for policymakers, as they should help them to evaluate if measures undertaken to reduce the size of informal economy are effective. However, one should not forget the limitations of the Currency Demand Approach. Future research should therefore not only focus on providing estimates of the size of informal economy for individual countries but also should aim at improving the methods used to obtain these estimates.

Notes

1. In this paper, we follow OECD (2002) in defining the informal economy as all legal production activities that are deliberately concealed from public authorities for the following kinds of reasons: to avoid payment of income, value added or other taxes; to avoid payment of social security contributions; to avoid having to meet certain legal standards such as minimum wages, maximum hours, safety or health standards, etc., to avoid complying with certain administrative procedures, such as completing statistical questionnaires or other administrative forms.
2. Formerly known as Central Statistical Office.
3. This functional form allows to interpret the coefficient as semi-elasticity of money demand to the amount of direct taxes scaled by GDP, interpretable as one possible measure of fiscalism level. No material changes to the results occur if the logarithm is not applied.
4. To avoid a level shift effect between these two subperiods, the registered data in the first subsample are augmented by 2.1 p.p. (which is the average difference between these two series in the first year when both are available, i.e. 2000).
5. Available at an annual frequency since 2002, earlier (1996–2002) bi-annually. In both cases, linear interpolation to quarterly data has been performed.
6. For details on standard error computation and distribution, see Pedroni (2000). The Engle–Granger cointegration test (with a constant and 4 lags) confirms this at the significance level of 0.05 with the p -value of 0.0313. By the Granger representation theorem, the estimates of the error-correction equation $d(y_t) = \alpha \varepsilon_{t-1} + \theta d(x_t) + \eta_t$ remain in line with this finding, yielding the error correction coefficient $\hat{\alpha} = -0.2277$, significant at the level of 0.01 (p -value 0.0014).
7. The second vector is just-identified as i.a. cash/M1 unaffected (i.e. excluded from the second cointegrating vector, $\beta_{2,1} = 0$, and not undergoing any alignment with the second relation, $\alpha_{2,1} = 0$, and normalized with respect to direct tax revenues, $\beta_{2,2} = 1$). Since it remains

unrelated to the CDA method, we leave its discussion outside the scope of this paper. For the results of the second vector please see [Table A3](#) in Appendix 1.

8. Note that this also corresponds to the estimates of Non-Accelerating Inflation Rate of Unemployment (NAIRU) for the USA, ranging from 4.7 to 5.9% according to Federal Reserve Bank of Philadelphia (<https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/nairu-data-set>). The use of NAIRU estimates for Poland might be potentially misleading in the CDA framework due to the high instability of the measurement results over the sample period.

Acknowledgements

The authors would like to thank two anonymous referees, the Editor, Karsten Staehr, Gilbert Mbara and the participants of the 7th Shadow Conference: The Shadow Economy, Tax Behaviour and Institutions (shadow2021), especially Jakub Sawulski as a discussant, for useful comments. The usual disclaimer applies.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendices

Appendix 1. Extended estimation results

Table A1. Correction coefficient estimates (α) in VECM-1 model.

i (variable)	DLOG (cash/M1)	DLOG (direct tax revenues)	D (unemployment rate)	D (gov. effectiveness)	DLOG (electronic payments)	D (inflation)	D (nominal interest rate)
$\hat{\alpha}_{1,i}$ (coint. eq. 1)	–0.304537 (0.11458) [–2.65790]				0.194809 (0.04754) [4.09750]	–16.64689 (3.24959) [–5.12277]	–7.030531 (1.26718) [–5.54818]
$\hat{\alpha}_{2,i}$ (coint. eq. 2)		–0.402294 (0.09762) [–4.12106]			0.117947 (0.01925) [6.12676]		

LR test for binding restrictions (rank = 2): Chi-square(9) 6.854845 Probability 0.652229. Source: authors' elaboration.

Table A2. Correction coefficient estimates (α) in VECM-2 model.

i (variable)	DLOG (cash/M1)	DLOG (direct tax revenues)	D (unemployment rate)	D (gov. effectiveness)	DLOG (electronic payments)	D (inflation)	D (nominal interest rate)
$\hat{\alpha}_{1,i}$ (coint. eq. 1)	–0.677621 (0.09456) [–7.16576]						
$\hat{\alpha}_{2,i}$ (coint. eq. 2)		–0.844391 (0.13974) [–6.04248]			0.126931 (0.03484) [3.64369]		

LR test for binding restrictions (rank = 2): Chi-square(11) 59.43572 Probability 0.000000.

Source: authors' elaboration.

Table A3. Currency demand model – VECM-1 second cointegrating vector estimates and VECM-2 second cointegrating vector estimates (1999q4-2019q4).

Variable	VECM-1 model estimates			VECM-2 model estimates		
	Estimate	Std. Error	t-Statistic	Estimate	Std. Error	t-Statistic
Direct tax revenues	1.000			1.000		
Unemployment rate	0.000			0.000		
Gov. Effectiveness	0.554	0.182	3.035	0.761	0.106	7.149
Electronic payments	−1.510	0.198	−7.644	−0.721	0.061	−11.754
Inflation	0.052	0.013	3.958	0.065	0.011	6.118
Nominal interest rate	−0.096	0.012	−8.012	−0.092	0.010	−9.262
Intercept	1.023			2.271		
Cash to M1 (log)	0.000			0.000		
Linear trend	0.010	0.003	3.568			

Source: authors' elaboration.

Table A4. Cointegration test results (trace and maximum eigenvalue).

Hypothesized number of cointegrating equations	Eigenvalue	Trace test – <i>p</i> -value	Maximum Eigenvalue test – <i>p</i> -value
0	0.580979	0.0000	0.0005
1	0.533921	0.0000	0.0008
2	0.388382	0.0101	0.0566
3	0.264647	0.1066	0.3709
4	0.220735	0.2047	0.2918
5	0.144398	0.4271	0.4144
6	0.060812	0.6208	0.6208

Source: authors' elaboration.

Appendix 2. Sensitivity check – dependent variable

Table A5. Currency demand model – FM-OLS, OLS, VECM-1 model ML and VECM-2 model ML estimates (1999q4-2019q4) with cash in circulation/M2 ratio as dependent variable.

Variable	FM-OLS estimates				OLS estimates				VECM-1 model estimates			VECM-2 model estimates		
	Coefficient	Std. Error	t-Statistic	Prob.	Coefficient	Std. Error	t-Statistic	Prob.	Estimate	Std. Error	t-Statistic	Estimate	Std. Error	t-Statistic
Direct tax revenues	0.213	0.110	1.934	0.057	0.269	0.086	3.115	0.003	0.602	0.120	5.023	0.659	0.187	3.521
Unemployment rate	0.014	0.008	1.874	0.065	0.011	0.006	1.864	0.067	0.011	0.004	3.018	0.022	0.005	4.015
Gov. effectiveness	−0.523	0.114	−4.603	0.000	−0.408	0.089	−4.602	0.000	−0.646	0.115	−5.622	−0.295	0.182	−1.619
Electronic payments	−0.324	0.188	−1.721	0.090	0.256	0.147	1.737	0.087	0.181	0.094	1.925	0.383	0.148	2.586
Inflation	−0.010	0.009	−1.161	0.249	−0.007	0.007	−0.959	0.341	0.018	0.009	2.069	−0.018	0.014	1.330
Nominal interest rate	−0.011	0.011	−0.979	0.331	−0.006	0.009	−0.675	0.502	−0.061	0.009	−6.467	−0.044	0.015	−2.949
Intercept	−0.807	0.337	−2.389	0.020	−0.767	0.264	−2.908	0.005	0.576			0.535		
Linear trend									0.000					

Note 1: For FM-OLS estimates – R -squared = 0.480, Adjusted R -squared = 0.436, S.E. of regression = 0.064. Long-run covariance estimate: prewhitening with 4 lags (selected with AIC under max. 4 lags), Bartlett kernel and Newey–West fixed bandwidth = 4. Note 2: For OLS estimates – R -squared = 0.637, Adjusted R -squared = 0.606, F -statistic = 20.491, Prob. (F -statistic) = 0.000. Source: authors' elaboration.

Table A6. Currency demand model – VECM-1 second cointegrating vector estimates and VECM-2 second cointegrating (1999q4-2019q4) vector estimates with cash in circulation/M2 ratio as dependent variable.

Variable	VECM-1 model estimates			VECM-2 model estimates		
	Estimate	Std. Error	t-Statistic	Estimate	Std. Error	t-Statistic
Direct tax revenues	1.000			1.000		
Unemployment rate	0.000			0.000		
Gov. effectiveness	0.368	0.135	2.728	0.720	0.098	7.356
Electronic payments	−1.814	0.201	−9.035	−0.711	0.056	−12.756
Inflation	0.037	0.011	3.450	0.067	0.010	6.981
Nominal interest rate	−0.099	0.009	−11.682	−0.096	0.009	−10.858
Intercept	0.596			2.318		
Cash to M2 (log)	0.000			0.000		
Linear trend	0.015	0.003	5.223			

Source: authors' elaboration.

Appendix 3. Additional graphs and tables

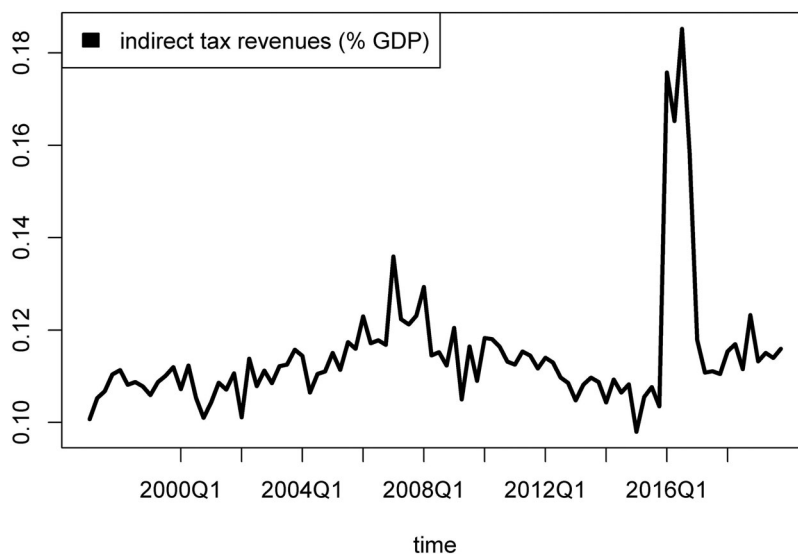


Figure A1. Indirect tax revenues in Poland, 1997–2018. Source: Statistical Bulletin, Statistics Poland.

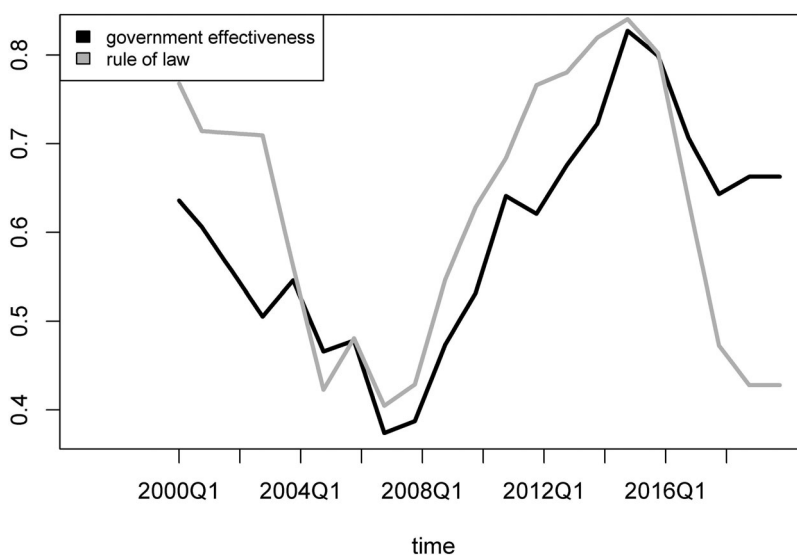


Figure A2. Government effectiveness index versus Rule of law index in Poland, 1997–2018. Source: World Bank.

Table A7. Unit root tests for the CDA equation variables.

Variable	ADF	ADF	KPSS statistic (level)	KPSS statistic (difference)
	<i>p</i> -value (level)	<i>p</i> -value (difference)		
Log (cash in circulation to M1)	0.644	0.000	3.033	0.106
Log (direct tax revenues)	0.084	0.000	0.406	0.020
Unemployment rate	0.320	0.372	2.736	0.355
Government effectiveness	0.326	0.044	1.772	0.423
Log (electronic payments)	0.499	0.000	2.967	0.707
Nominal interest rate	0.100	0.000	1.064	0.130
Inflation	0.304	0.000	2.131	0.151

KPSS critical values: 0.739 (sig. level 0.01), 0.463 (sig. level 0.05), 0.347 (sig. level 0.10). None of the variables is trendstationary. Source: authors' elaboration.

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